MILITARY SPECIFICATION

COLLIMATOR, INFINITY AIMING REFERENCE: XM2

1. SCOPE

1.1 This specification covers an infinity aiming reference collimator which consists of a collimator assembly, tripod, cable assembly, and strap. The complete instrument is referenced throughout this specification as "collimator" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1240

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DRAWINGS

U. S. Army Munitions Command

F10533251	Collimator, Infinity Aiming Reference: XM2
IEL 10533251	List of Inspection Documents for Collimator, Infinity Aiming Reference: XM2

PACKAGING DATA SHEET

MIL-P-14232/P5801010	Packaging of Collimator, Infinity Aiming Reference: XM2 W/Case

(Copies of specifications, standards, drawings, and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The collimator shall be manufactured in accordance with Drawing F10533251 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Part identification and marking
(f) Workmanship
(g) Soldering electrical connections

3.3 Environmental service conditions.

3.3.1 Storage temperatures.- The collimator shall show no evidence of physical failure and shall meet the requirements of 3.4 to 3.9.5 inclusive at a standard ambient temperature between +60° and +90° Fahrenheit (F) after having been exposed to and thermally stabilized at ambient temperatures of 80° and +160°F.

3.3.2 Extreme operating temperatures.- The collimator shall meet the requirements of 3.7.1 to 3.8.6 inclusive and 3.9.3 while exposed to and thermally stabilized at ambient temperatures of -65 and +150°F.

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3.3.3 Vibration.- The collimator shall show no evidence of physical failure and shall meet the requirements of 3.4 to 3.9.5 inclusive after being vibrated at a frequency of 30 ± 5 cycles per second and an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds.

3.4 Cleanliness.- The optical system shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than 3 particles of dirt visible on the reticle. The area of each particle shall not be in excess of the area equivalent to a size 10 dig. There shall be no other dirt within the optical system which interferes with observation or adversely affects image quality.

3.5 Sealing.- The tube assembly of the collimator shall show no evidence of leakage when subjected to an internal pressure of dry nitrogen of 5 ± 0.10 pounds per square inch gage (psig) for 5 minutes (minimum). The dew point of the nitrogen shall be at least as low as -25°F.

3.6 Accuracy.

3.6.1 Reticle lean.- With the zero vertical reticle line parallel to the image of a plumb line within 5 minutes of arc, the level vial bubble shall be centered within 0.1 of a division.

3.6.2 Reticle accuracy.- The angle subtended by the two 50 mil graduation lines on the horizontal axis of the reticle pattern shall be 100 mils ± 0.1 mil.

3.6.3 Parallax.- Using a telescope having a one inch entrance pupil, and with the reticle of the telescope adjusted for infinity focus, the parallax between that reticle and the center cross of the collimator reticle shall not exceed 0.1 mil.

3.7 Mechanical.

3.7.1 Elevation travel.- The collimator shall provide elevation of the line of sight from -800 mils to +800 mils when the tripod legs are fully extended and resting on a horizontal surface.

3.7.2 Azimuth travel.- The collimator shall provide unrestricted azimuth rotation of 6400 mils in either direction.

3.8 Operability.- There shall be no looseness of assembly, restriction, or binding of the equipment when the collimator is placed in the operational position. The tripod leg assemblies shall be assembled to provide an operating torque of 5 pound-feet.

3.8.1 Knob and lever travel.- Clockwise rotation of the collimator clamping knob, elevation clamping lever, and azimuth clamping lever shall lock applicable components in position. Counterclockwise rotation of the knob and levers shall release the components for free movement.

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3.8.2 Knob and lever locks.- With 35 pound-inches of torque used to secure the locking mechanisms, the parts which the locking mechanisms are intended to hold shall not move when the torque values specified below are applied.

Locking control for	Torque (min)
(a) Rotating collimator in azimuth on tripod	50 pound-inches
(b) Cross leveling the collimator	50 pound-inches
(c) Elevating the collimator	8 pound-feet

3.8.3 Scope alinement.- With the lock released, the torque required to rotate the scope (collimator assembly) to center the level vial bubble shall not be less than 5 pound-inches nor more than 15 pound-inches at a standard ambient temperature between +60° and +90°F. The maximum torque shall not exceed 25 pound-inches at the extreme temperatures of 3.3.2.

3.8.4 Switches and potentiometer knob.- The toggle switches, pushbutton switch, and potentiometer knob shall be mechanically operable when the collimator is exposed to the extreme temperatures of 3.3.2.

3.8.5 Replacement of batteries and lamp.- The draw-pull compression catch and end housing cover shall be operable, and the batteries and lamp shall be removable and replaceable at the extreme operating temperatures of 3.3.2.

3.8.6 Level vial cover.- The level vial cover shall be operable at the extreme temperatures of 3.3.2.

3.9 Electrical.

3.9.1 Reticle illumination.- With the reticle illuminated, the reticle pattern shall appear clearly defined when observed in a dark room through a four power telescope. Operation of the potentiometer knob in a clockwise direction shall increase the intensity of illumination; counterclockwise rotation of the knob shall decrease the light intensity.

3.9.2. . Circuitry.- The circuitry of the collimator shall be in accordance with the applicable electrical schematic drawing.

3.9.3 Cable assembly.- The cable assembly shall be operable and the cable shall show no evidence of physical failure at the extreme operating temperatures of 3.3.2.

3.9.4 Dielectric strength.- There shall be no puncture of insulation, no spark-over, nor decrease in dielectric strength, when the insulated areas of the circuitry is subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

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3.9.5 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with a direct current (dc) potential of between 200 and 500 volts applied for 1 minute (minimum).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 10533251. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

CRITICAL: NONE

MAJOR: AQL .65 Percent defective	REQUIREMENT	TEST PROCEDURE
101. Physical failure, post vibration	3.3.3	4.5.1.2
102. Cleanliness	3.4	4.5.2
103. Reticle lean	3.6.1	4.5.4.1
104. Reticle accuracy	3.6.2	4.5.4.2
105. Parallax	3.6.3	4.5.4.3
MAJOR: AQL 1.0 percent defective		
106. Elevation travel	3.7.1	4.5.5.1
107. Azimuth travel	3.7.2	4.5.5.2
108. Operability (60° to 90°F)	3.8	4.5.6.1
109. Knob and lever travel	3.8.1	4.5.6.2
110. Knob and lever locks	3.8.2	4.5.6.3
111. Torque, scope alinement	3.8.3	4.5.6.4
112. Switches and potentiometer knob	3.8.4	4.5.6.5
113. Batteries and lamp	3.8.5	4.5.6.6
114. Level vial cover	3.8.6	4.5.6.7
115. Reticle illumination	3.9.1	4.5.7.1
116. Circuitry	3.9.2	4.5.7.2
117. Cable assembly	3.9.3	4.5.7.3
118. Dielectric strength	3.9.4	4.5.7.4
119. Insulation resistance	3.9.5	4.5.7.5

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests.

4.3.1 General.- One collimator shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings
302. General specification	3.2	Visual-MIL-F-13926

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4.3.2 Environmental.- Three collimators shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Temperature tests	3.3.1 & 3.3.2	4.5.1.1, 4.5.5 & 4.5.6
304. Sealing test	3.5	4.5.3

4.3.3 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL 10533251.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 Devices used for environmental testing.- All environmental testing devices used to determine conformance of the environmental requirements (listed below) shall conform to the "Test Facilities" requirements outlined in MIL-F-13926.

(a) Vibration (3.3.3)
(b) Temperature (3.3.1) (3.3.2)
(c) Sealing (3.5)

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4.4.2.4 Device to test reticle lean, reticle accuracy, parallax and elevation travel.- This device shall be so constructed as to permit location of the collimator in its normal operating position and shall have sufficient mediums to test these requirements. The mediums noted below shall be to the accuracies stated. Each medium used for measuring reticle lean, reticle accuracy, parallax and elevation travel shall have sufficient acceptance and rejection limitation markings or pips marked thereon and shall be within ten (10) percent of the specified tolerance and plumb within twenty-five (25) seconds.

(a) Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are prime requirements.

(b) All special set-up and calibration equipment necessary for certification of device accuracy shall be furnished and will become a part of the device.

4.5 Test methods and procedures.

4.5.1 Environmental tests.

4.5.1.1 Storage and operating temperatures.- This test shall be conducted in accordance with Procedure I of MIL-F-13926. While stabilized at each temperature extreme specified in 3.3.2 the collimator shall be subjected to the tests specified in 4.5.5 through 4.5.6.7 inclusive, and shall conform to the requirements specified therein. Upon completion of the specified exposures and return to standard ambient temperature (60° to 90°F) the collimator shall be inspected and shall show no evidence of glass breakage or other physical failure in accordance with 3.3.1. The collimator shall then be subjected to all the tests in this specification and shall meet the requirements specified.

4.5.1.2 Vibration.- This test shall be conducted at a standard ambient temperature between +60° and +90°F in accordance with Procedure V of MIL-F-13926, except that the collimator assembly shall be secured in a horizontal position and the time shall be 5 minutes ± 15 seconds. Subsequent to vibration the collimator shall be subjected to a visual and tactile examination and shall show no evidence of damage. The collimator shall then be subjected to the remaining tests in this specification and shall meet the specified requirements.

4.5.2 Cleanliness.- Immediately after vibration the optical system of the collimator shall be inspected for cleanliness in accordance with MIL-O-13830 and 3.4 herein and shall meet the requirements specified. Dirt particles which interfere with the legibility or accuracy of the reticle pattern shall be cause for rejection. The mean diameter of a size 10 dig is equal to 0.10 millimeter.

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4.5.3 Sealing test.- The sealing tests shall be accomplished by means of commercial equipment which shall include a sensitive pressure gage capable of testing to the requirements of 3.5. The equipment shall be attached to the collimator by means of the #8-32 threaded hole in the tube. The pressure of dry nitrogen gas indicated in 3.5 is applied and then cut off. Evidence of leakage can be determined by observing the gage for drop of pressure for time interval specified. Upon completion of the test, the collimator shall be bled to the ambient atmospheric pressure and sealed.

4.5.4 Accuracy tests.

4.5.4.1 Reticle lean.- Reticle lean test shall be performed with the aid of an auxiliary telescope set to the proper eyepiece focus. The vertical reticle line of the telescope shall be parallel to a plumb line and adjusted to infinity focus. Reticle accuracies shall meet the requirements of 4.4.2.4. Position the collimator in front of the auxiliary telescope so the optical axes are coincident. Align the zero vertical line image of the collimator parallel with the vertical reticle line of the auxiliary telescope. When the zero vertical line image of the collimator is plumbed, the bubble of the level vial shall be within the central graduation marks of the vial in accordance with the requirements of 3.6.1.

4.5.4.2 Reticle accuracy.- This test shall be accomplished with the aid of an auxiliary telescope and calibrated reticle with accuracies as specified in 4.4.2.4. The calibrated reticle shall have a horizontal line 100 mils in length with vertical marks at the center and each end. A mil block at either end of the horizontal line will enable the observer to detect error. With the auxiliary telescope focused on the collimator reticle image, both 50 mil marks of the reticle image shall coincide within the allowable tolerance in accordance with the requirements of 3.6.2.

4.5.4.3 Parallax.- This test shall be performed with the auxiliary telescope positioned as specified in 4.5.4.1. Parallax shall be removed from the auxiliary telescope by means of auto collimation. The test shall be made by sighting through the auxiliary telescope to the collimator center cross image. Parallax will be recognized as any apparent displacement of the collimator center cross image in relation to the auxiliary telescope reticle when the observer's head is moved from side to side or up and down. Any movement of the observer's head shall not cause displacement of the collimator center cross image to exceed the requirement specified in 3.6.3.

4.5.5 Mechanical tests.

4.5.5.1 Elevation travel.- The elevation travel test shall be made by means of a wall chart located a suitable distance from the collimator. The chart shall have a plumbed line with a mark for the line of sight of the collimator and marks located 800 mils above and below the collimator line of sight mark. With legs fully extended in a horizontal position and the elevation clamping mechanism released, the collimator shall be elevated and depressed through the entire excursion indicated. Sighting of the collimator shall be accomplished by attaching the auxiliary telescope to the collimator, which shall meet the minimum requirements specified in 3.7.1.

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4.5.5.2 Azimuth travel.- This test shall be made with the wall chart specified in 4.5.5.1. Sighting of the collimator shall be accomplished by attaching the auxiliary telescope to the collimator. With the collimator aligned with the plumbed line and the azimuth clamping mechanism released, the collimator shall be rotated 6400 mils and shall meet the requirements of 3.7.2.

4.5.6 Operability tests.

4.5.6.1 Operability.- The collimator shall be manually set up in the operational position from the carrying position. This shall be a visual and tactile examination of the collimator for conformance to the requirements specified in 3.8.

4.5.6.2 Knob and lever travel.- There shall be a tactile examination for movement of the knob and levers to provide full locking and free movement of each specific component as specified in 3.8.1.

4.5.6.3 Knob and lever locks.- With each knob and lever of the collimator locked by application of the measured torque value in 3.8.2, there shall be no movement of the applicable collimator components when the specified test values are applied in conformance with the requirements of 3.8.2. The measured torque values shall be applied with standard measuring equipment.

4.5.6.4 Scope alignment.- The torque required to rotate the scope (collimator assembly) to center the level vial shall be measured with standard measuring equipment. The measured torque values shall not exceed the values specified in 3.8.3 for the temperatures indicated.

4.5.6.5 Switches and potentiometer knob.- This test shall be accomplished by visual and tactile means to the conditions specified in 3.8.4.

4.5.6.6 Batteries and lamp.- The end housing cover catch shall be operated and the batteries and lamp removed by normal manual means when exposed to the temperatures specified in 3.8.5.

4.5.6.7 Level vial cover.- The cover shall be operated by normal means when exposed to the temperatures specified in 3.8.6.

4.5.7 Electrical tests.

4.5.7.1 Reticle illumination.- The reticle illumination test shall be accomplished by means of a visual examination in a darkened area with the reticle lamp energized by a 3-volt source. The intensity of the lamp shall be varied from maximum to minimum illumination. The reticle markings and light intensity shall be clearly defined when observed in accordance with the requirements of 3.9.1.

4.5.7.2 Circuitry.- A continuity check of the collimator shall be made utilizing standard testing equipment to determine conformance with the requirements specified in 3.9.2.

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4.5.7.3 Cable assembly.- When exposed to the temperatures specified in 3.9.3 the switch shall be operated manually. Subsequent to the specified exposure, the cable and switch shall be inspected by visual and tactile means for any evidence of failure.

4.5.7.4 Dielectric strength.- This test shall be performed in accordance with Dielectric Strength, Method 301 of MIL-STD-202. Standard measuring equipment shall be used for all measurements. There shall be no evidence of failure or decrease in dielectric strength when subjected to the specified voltage for the time period indicated in 3.9.4.

4.5.7.5 Insulation resistance.- Insulation resistance shall be measured using Method 302, Test Condition B, of MIL-STD-202, by means of standard measuring equipment. Resistance shall be measured at areas specified and shall meet the minimum resistance specified in conformance with the requirements of 3.9.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P5801010. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Infinity Aiming Reference Collimator XM2 is a portable optical instrument intended to be used for indirect laying of weapons (primarily mortars) by establishing an optical reference line from which weapon deflection angles can be measured.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

Custodian:

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Preparing activity:

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